

ARTIFICIAL INTELLIGENCE

UNIT – I

Short Answer Questions + Long Answer Questions

Exam-oriented answers with clear explanations and examples

Course	Artificial Intelligence
Unit	Unit – I
Short Answers	10 Questions – 2 Marks each
Long Answers	Questions 1.a to 10.b – 5/10 Marks

How to use: For 2-mark questions, write the definition plus key points/example. For 5-mark questions, write the definition, main points, example, and conclusion. For the 10-mark comparison question, include all major environment dimensions and examples.

UNIT – I: SHORT ANSWER QUESTIONS

1. Define Artificial Intelligence. Mention any two characteristics of an AI system. [2M]

Artificial Intelligence (AI) is the branch of computer science that develops systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, perception, decision-making, and language understanding.

Two characteristics:

- **Learning:** The system can learn from data or experience and improve its performance.
- **Reasoning/decision-making:** The system can analyze information and choose suitable actions to achieve a goal.

2. List any four application areas of Artificial Intelligence. [2M]

- **Healthcare:** disease prediction, medical image analysis, and drug discovery.
- **Finance:** fraud detection, credit assessment, and algorithmic analysis.
- **Transportation:** route planning, traffic prediction, and autonomous vehicles.
- **Education:** intelligent tutoring systems, personalized learning, and automatic assessment.

3. What is the PEAS model? Illustrate it with a simple agent example. [2M]

PEAS stands for **Performance measure, Environment, Actuators, and Sensors**. It is used to specify an intelligent agent's task environment.

Example – Vacuum-cleaner agent:

- **Performance:** Clean floor, minimum time/energy, avoid unnecessary movement.
- **Environment:** Rooms containing clean or dirty floor areas.
- **Actuators:** Move left/right, suck dirt.
- **Sensors:** Dirt sensor and location sensor.

4. Mention any four important milestones in the history of Artificial Intelligence. [2M]

- **1950:** Alan Turing proposed the idea behind the Turing Test.
- **1956:** The Dartmouth workshop helped establish AI as a formal research field.
- **1997:** IBM's Deep Blue defeated world chess champion Garry Kasparov.
- **2012:** Deep learning achieved a major breakthrough in image recognition, accelerating modern AI research.

5. Define an Intelligent Agent. [2M]

An **intelligent agent** is an entity that **perceives its environment through sensors and acts upon that environment through actuators** to achieve goals. It selects actions that are appropriate for the situation.

Example: A self-driving car uses cameras and other sensors to perceive roads and controls steering and speed through its actuators.

6. What is the difference between an agent and its environment? [2M]

An **agent** is the system that perceives and acts, while the **environment** is everything external to the agent that it interacts with.

Example: In a vacuum-cleaner problem, the vacuum cleaner is the agent; the rooms, dirt, walls, and floor are part of the environment.

Relationship: Environment → sensors → Agent → actuators → Environment.

7. What is meant by a rational agent? [2M]

A **rational agent** is an agent that chooses the action expected to **maximize its performance measure**, based on its percept sequence, available knowledge, and possible actions.

Rationality does not mean that the agent is always successful; it means it chooses the best action that can reasonably be determined from the information available.

8. Classify environments based on their characteristics in Artificial Intelligence. [2M]

AI task environments are commonly classified using these properties:

- **Fully observable / partially observable** – whether the agent can perceive the complete relevant state.
- **Deterministic / stochastic** – whether an action has a predictable result.
- **Episodic / sequential** – whether decisions depend on previous actions.
- **Static / dynamic** – whether the environment changes while the agent is deciding.
- **Discrete / continuous** – whether states/actions occur in countable or continuous forms.
- **Single-agent / multi-agent** – whether one or multiple agents interact.

9. What is a problem-solving agent? [2M]

A **problem-solving agent** is a goal-based intelligent agent that formulates a problem and searches for a sequence of actions that reaches a desired goal state.

Basic process: Goal formulation → problem formulation → search → action execution.

Example: A route-planning agent searches for a path from a starting city to a destination city.

10. Define problem formulation in Artificial Intelligence. What are its main components? [2M]

Problem formulation is the process of representing a real-world goal as a formal search problem so that an AI system can find a solution.

Main components:

- Initial state
- Actions / successor function
- State space
- Goal test
- Path cost

UNIT – I: LONG ANSWER QUESTIONS

1.a. Explain the concept of Artificial Intelligence and discuss its characteristics and applications. [5M]

Meaning: Artificial Intelligence is the field of computer science concerned with building systems that can perceive information, reason about it, learn from experience, and take actions to achieve goals.

Characteristics of AI:

- **Learning:** AI systems can learn patterns from data and experience.
- **Reasoning:** They can draw conclusions from available facts.
- **Problem solving:** They can search for solutions to complex problems.
- **Perception:** They can process images, speech, and other sensor information.
- **Language understanding:** They can process and generate human language.
- **Decision-making:** They can select actions according to goals and constraints.

Applications: AI is used in healthcare for diagnosis support, finance for fraud detection, transportation for route planning, education for personalized learning, agriculture for crop monitoring, cybersecurity for anomaly detection, and entertainment for recommendation systems.

Conclusion: AI combines data, algorithms, and computing to create systems that perform intelligent tasks and support human decision-making.

1.b. Describe the foundations of Artificial Intelligence and explain how different disciplines contribute to AI. [5M]

AI developed from ideas contributed by several disciplines:

- **Philosophy:** contributes logic, reasoning, knowledge, and questions about intelligent behavior.
- **Mathematics:** provides logic, probability, statistics, optimization, and linear algebra.
- **Economics:** contributes utility, decision theory, and models of rational choice.
- **Neuroscience:** provides inspiration about how biological brains process information.
- **Psychology:** contributes models of perception, learning, memory, and problem solving.
- **Computer science:** provides algorithms, programming, data structures, and computational methods.
- **Control theory:** contributes feedback, stability, and decision-making for dynamic systems.
- **Linguistics:** contributes theories and structures for language processing.

Conclusion: AI is interdisciplinary. Modern intelligent systems combine mathematical methods, computing, learning, perception, language, and decision theory.

2.a. Discuss the history and evolution of Artificial Intelligence, highlighting major milestones. [5M]

Early foundations: Before AI became a formal field, logic, mathematics, computing, and theories of intelligent behavior provided its foundation.

- **1950 – Turing Test:** Alan Turing proposed a test for machine intelligence based on whether a machine could communicate in a way indistinguishable from a human.
- **1956 – Dartmouth workshop:** The term Artificial Intelligence became associated with a formal research field.
- **1960s–1970s:** Researchers developed symbolic reasoning, search methods, early natural-language programs, and robotics.

- **1970s–1980s:** Expert systems became important because they encoded specialist knowledge as rules.
- **AI winters:** Periods of reduced funding and enthusiasm occurred when systems failed to meet expectations.
- **1997 – Deep Blue:** IBM's chess system defeated Garry Kasparov.
- **2010s – Deep learning:** Large datasets, GPUs, and improved neural networks produced major progress in vision and speech.
- **2020s – Generative AI:** Large foundation models enabled powerful text, image, audio, and multimodal applications.

Conclusion: AI evolved from rule-based symbolic systems toward data-driven, learning-based and foundation-model systems.

2.b. Analyze the impact of important AI developments on modern intelligent systems. [5M]

Important developments changed AI in several ways:

- **Expert systems:** demonstrated that domain knowledge could be represented computationally for decision support.
- **Machine learning:** shifted the focus from manually written rules toward systems that learn patterns from data.
- **Deep learning:** improved image recognition, speech recognition, natural-language processing, and other perception tasks.
- **Reinforcement learning:** enabled agents to learn actions through rewards and interaction with environments.
- **Transformers and foundation models:** greatly improved language understanding, generation, and multimodal capabilities.
- **Cloud computing and GPUs:** made large-scale model training and deployment more practical.

Overall impact: Modern systems are more adaptive, capable, and scalable. At the same time, issues such as bias, privacy, reliability, transparency, and computational cost have become important considerations.

3.a. Explain the concept of Intelligent Agents with suitable examples. [5M]

Definition: An intelligent agent perceives its environment through sensors and acts through actuators to achieve a goal or maximize a performance measure.

Agent cycle:

- Environment produces a situation.
- Sensors receive percepts.
- The agent interprets the percepts and selects an action.
- Actuators perform the action in the environment.
- The process repeats as new percepts arrive.

Examples:

- **Robot vacuum:** sensors detect dirt and obstacles; motors and suction perform actions.
- **Self-driving car:** cameras and other sensors detect road conditions; steering, braking, and acceleration act on the vehicle.
- **Recommendation system:** user interactions are percepts and recommended content is an action.

Key point: An agent is intelligent when its behavior appropriately uses information and goals to select actions.

3.b. Describe the interaction between agents and environments using the PEAS framework. [5M]

PEAS describes the task environment of an agent using four elements:

- **P – Performance measure:** criteria used to judge how well the agent performs.
- **E – Environment:** the external world in which the agent operates.
- **A – Actuators:** mechanisms used to perform actions.
- **S – Sensors:** mechanisms used to obtain percepts.

Example – Self-driving car:

- **Performance:** safety, reaching the destination, obeying traffic rules, comfort, and efficiency.
- **Environment:** roads, traffic lights, pedestrians, other vehicles, weather, and road conditions.
- **Actuators:** steering, brakes, accelerator, and signals.
- **Sensors:** cameras, radar, GPS, and other vehicle sensors.

Interaction: Sensors collect information → agent processes percepts → agent selects an action → actuators affect the environment → new percepts are received.

4.a. Explain the concept of rationality in intelligent agents. [5M]

Rationality means selecting the action that is expected to produce the best outcome according to the agent's performance measure, given its percept history, knowledge, and available actions.

Factors affecting rational action:

- **Performance measure:** defines what counts as success.
- **Percept sequence:** the information received so far.
- **Prior knowledge:** information the agent already has about the environment.
- **Available actions:** the choices the agent can make.
- **Uncertainty:** possible outcomes may not be completely predictable.

Example: A route-planning agent is rational if it chooses a route expected to provide a good balance of travel time, safety, and other specified performance criteria.

Important: Rationality is not the same as perfection. A rational agent can make an unsuccessful choice when information is incomplete or the environment is uncertain.

4.b. Analyze the factors that influence the performance of a rational agent in different environments. [5M]

The performance of a rational agent depends on several environmental and agent-related factors:

- **Observability:** Full information makes decision-making easier; partial observability requires estimation.
- **Determinism:** Predictable outcomes simplify planning; stochastic outcomes require reasoning under uncertainty.
- **Number of agents:** Other agents may cooperate, compete, or create unpredictable changes.
- **Static or dynamic nature:** A dynamic environment may change while the agent is thinking.
- **Time and computation:** Limited resources can prevent exhaustive reasoning.
- **Knowledge and learning:** Better models and experience can improve future decisions.
- **Sensor/actuator limitations:** Noisy sensors or restricted actions reduce achievable performance.

Conclusion: A rational agent must adapt its decision strategy to the properties of the environment and the resources available.

5.a. Explain the characteristics of task environments in Artificial Intelligence. [5M]

Task environments are described using important dimensions:

- **Fully observable vs partially observable:** In a fully observable environment the agent can access all relevant information; in a partially observable environment some information is hidden or noisy.
- **Deterministic vs stochastic:** A deterministic action has a predictable outcome; a stochastic environment involves uncertainty.
- **Episodic vs sequential:** In episodic tasks, each decision is relatively independent; in sequential tasks, current actions affect future situations.
- **Static vs dynamic:** Static environments do not change while the agent deliberates; dynamic environments can change.
- **Discrete vs continuous:** Discrete environments have countable states/actions/time steps; continuous environments involve continuous values.
- **Single-agent vs multi-agent:** Only one agent acts in a single-agent task; multiple interacting agents exist in a multi-agent task.

Example: Chess is largely fully observable, deterministic, sequential, discrete, and multi-agent, whereas driving is partially observable, stochastic, sequential, dynamic, continuous, and multi-agent.

5.b. Explain the structure of intelligent agents with a neat block diagram. [5M]

Structure: An intelligent agent consists of sensors, an agent program running on a computational architecture, and actuators.

Basic block diagram:

ENVIRONMENT → SENSORS → AGENT PROGRAM / ARCHITECTURE → ACTUATORS → ENVIRONMENT

- **Sensors:** collect percepts from the environment.
- **Agent program:** maps percept sequences to actions.
- **Architecture:** the hardware/software platform on which the agent program runs.
- **Actuators:** execute the selected actions.

Agent function: It can be represented as a mapping from percept sequences to actions: $f : P^* \rightarrow A$, where P^* is the set of percept sequences and A is the set of actions.

Example: In a robot, cameras and sensors provide percepts, the controller decides what to do, and motors perform movement.

6.a. Compare Simple Reflex, Model-Based Reflex, Goal-Based, Utility-Based, and Learning Agents. [5M]

1. **Simple Reflex Agent:** Uses condition–action rules based only on the current percept. It works well in simple, fully observable environments but cannot handle hidden information well.
2. **Model-Based Reflex Agent:** Maintains an internal model/state to handle partially observable environments. It uses current percepts plus internal information.
3. **Goal-Based Agent:** Selects actions by considering goals and often uses search/planning to find suitable action sequences.
4. **Utility-Based Agent:** Uses a utility function to compare outcomes and choose an action that gives the highest expected utility, especially when several goals or trade-offs exist.
5. **Learning Agent:** Improves its behavior from experience or feedback. A common architecture includes a performance element, learning element, critic, and problem generator.

Comparison: Simple reflex is the least flexible; model-based adds internal state; goal-based adds explicit goals; utility-based handles preferences and trade-offs; learning agents can improve from experience.

6.b. Explain the working of a problem-solving agent. [5M]

A problem-solving agent is a goal-based agent that searches for a sequence of actions leading from an initial state to a goal state.

Working steps:

- **1. Goal formulation:** Decide what desired state should be reached.
- **2. Problem formulation:** Define initial state, actions, transition model, goal test, and path cost.
- **3. Search:** Explore possible states using an appropriate search strategy such as BFS, DFS, or A*.
- **4. Solution:** Select an action sequence that reaches the goal.
- **5. Execution:** Execute the actions in the environment.
- **6. Replanning:** If the environment differs from expectations, the agent can update its plan.

Example: For travelling from City A to City B, the agent models cities as states, roads as actions/transitions, City B as the goal, and distance or time as path cost.

7.a. Illustrate the steps involved in problem formulation with a suitable real-world example. [5M]

Problem formulation converts a real-world objective into a formal search problem.

Example – Route planning:

- **Step 1: Identify the goal:** Reach City B from City A.
- **Step 2: Initial state:** The agent is at City A.
- **Step 3: Actions:** Travel along an available road to a neighboring city.
- **Step 4: State space:** All relevant cities and possible locations reachable through roads.
- **Step 5: Transition model:** Taking a road changes the current city to the connected city.
- **Step 6: Goal test:** Check whether the current city is City B.
- **Step 7: Path cost:** Total distance, travel time, fuel, or another chosen cost.

Result: A search algorithm can then find a suitable route from the initial state to the goal.

7.b. Explain the components of problem formulation in Artificial Intelligence. [5M]

The major components are:

- **Initial state:** The starting condition of the problem.
- **Actions:** The possible operations available in a state.
- **Transition model / successor function:** Describes the state produced by applying an action.
- **State space:** The set of states reachable through possible actions.
- **Goal test:** Determines whether a state satisfies the desired goal.
- **Path cost:** A numerical measure of the cost of an action sequence, often the sum of step costs.

Example: In a maze, the initial state is the starting cell, actions are moves, the state space is the reachable cells, the goal test checks the destination, and path cost may be the number of moves.

8.a. Formulate an AI problem by identifying the initial state, actions, state space, goal test, and path cost for a real-world scenario. [5M]

Scenario – Delivery robot finding a route inside a building:

- **Initial state:** Robot is at the warehouse/start location.
- **Actions:** Move north, south, east, or west where movement is permitted.
- **State space:** All accessible positions in the building together with relevant robot state.
- **Goal test:** Robot reaches the specified delivery location.
- **Path cost:** Total distance, time, energy, or a weighted combination.

Problem statement: Find a sequence of legal movements that takes the robot from its starting position to the delivery location while minimizing the selected path cost.

Note: Obstacles such as walls can be represented by disallowing the corresponding actions.

8.b. Discuss the role of intelligent agents in solving AI problems. [5M]

Intelligent agents provide a practical framework for applying AI to real-world tasks.

- **Perception:** Agents gather information through sensors or data sources.
- **Problem formulation:** Agents convert goals into formal problems with states and actions.
- **Reasoning and search:** Agents evaluate alternatives and find action sequences.
- **Decision-making:** Rational agents choose actions according to performance measures or utility.
- **Learning:** Learning agents improve from experience and feedback.
- **Interaction:** Agents continuously act in and respond to their environments.

Example: A navigation agent senses location and traffic information, searches for possible routes, chooses a suitable route, and updates the route as conditions change.

9.a. Compare the different types of environments with suitable examples. [10M]

1. Fully observable vs Partially observable: Chess is largely fully observable because the board is visible; driving is partially observable because some road conditions and future actions of others are unknown.

2. Deterministic vs Stochastic: A puzzle such as the 8-puzzle is deterministic; a taxi-driving environment is stochastic because traffic and other drivers introduce uncertainty.

3. Episodic vs Sequential: Image classification can be treated as episodic; chess is sequential because earlier moves affect later positions.

4. Static vs Dynamic: A crossword puzzle is static while solving; road traffic is dynamic because the environment changes continuously.

5. Discrete vs Continuous: Board games are discrete; robot movement and vehicle speed are continuous.

6. Single-agent vs Multi-agent: Solving a crossword alone is single-agent; chess and traffic involve multiple agents.

Summary: These properties determine which agent architecture, search method, planning strategy, and learning approach are suitable for a task.

10.a. Evaluate the suitability of different types of intelligent agents for real-world applications. [5M]

Simple Reflex Agents: Suitable for simple, predictable tasks such as basic automatic control. They are fast but limited because they rely on the current percept.

Model-Based Reflex Agents: Suitable when the environment is partially observable, such as a robot maintaining an internal representation of its surroundings.

Goal-Based Agents: Suitable for route planning, scheduling, puzzle solving, and other tasks where an explicit target must be reached.

Utility-Based Agents: Suitable for complex decisions involving trade-offs, such as choosing a route considering time, safety, and fuel cost.

Learning Agents: Suitable for changing environments and data-rich applications such as recommendation, speech recognition, adaptive robotics, and game playing.

Evaluation: No single agent type is best for every task. Real-world systems often combine multiple ideas—for example, a robot may use a model of its environment, goals, utility, and learning mechanisms together.

10.b. Explain the relationship between AI problems, intelligent agents, and problem formulation. [5M]

AI problem: A task that requires an intelligent system to reason, search, learn, perceive, or make decisions to achieve a desired outcome.

Intelligent agent: The entity that perceives the environment and performs actions to solve the task according to a performance measure.

Problem formulation: The process of representing the task in a structured form that an agent can solve. It specifies the initial state, actions, state space, goal test, and path cost.

Relationship:

- The **AI problem** defines what needs to be achieved.
- The **agent** is the decision-making system that attempts to achieve it.
- **Problem formulation** converts the real-world task into a searchable formal representation.
- A **search/planning method** finds a suitable action sequence.
- The agent **executes and evaluates** the solution in the environment.

Example: For navigation, reaching a destination is the AI problem; the navigation system is the agent; the current location, roads, possible movements, destination test, and travel cost form the problem formulation.

Conclusion: These three concepts are connected: **problem** → **formulation** → **agent solves and acts**.